



nextwork.org

VPC Endpoints



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vpce-00308dfd7afbfb29f / NextWork VPC Endpoint			
Details Route tables Policy Tags			
Details			
Endpoint ID vpce-00308dfd7afbfb29f	Status Available	Creation time Sunday, August 5, 2023 at 18:47:43 CDT	Endpoint type Gateway
VPC ID vpc-cf110214de9005865 (NextWork-vpc)	Status message -	Service name com.amazonaws.us-east-2.s3	Private DNS names enabled No
Service region us-east-2			



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a logically isolated space in the amazon cloudspace which helps us uniquely identify, manage, and route traffic to/from our resources in the cloudspace.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to launch and EC2 instance, setup a S3 bucket. Then, interact with the S3 bucket via the internet and via a VPC endpoint. Also, managed access via S3 bucket and endpoint policies.

One thing I didn't expect in this project was...

One thing I didn't expect in this project is the ability to manage granular access via policies.

This project took me...

This project took me about 90 minutes.



In the first part of my project...

Step 1 - Architecture set up

In this instance, we will create a VPC, launch an EC2 instance and create a S3 bucket.

Step 2 - Connect to EC2 instance

In this step, we are connecting directly to our EC2 instance to test if we can interact with our S3 bucket via the public internet.

Step 3 - Set up access keys

In this step, we are creating Access keys to give our EC2 instance access to our AWS environment.

Step 4 - Interact with S3 bucket

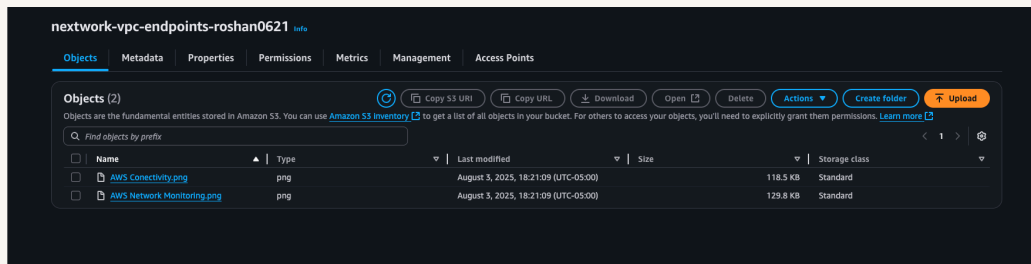
In this step, we are configuring our EC2 instance using the access key just created and setting up access to our AWS environment such that we can interact with the S3 bucket via the public internet.



Architecture set up

In this step, I created a VPC and launched an EC2 instance in the public subnet of the VPC.

I also set up a S3 bucket and uploaded two files to it.





Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, I configured the AWS access key ID, AWS Secret Access Key, Default region name, and Default output format.

Access keys are credentials for your applications and other servers to log into AWS and communicate with your AWS services/resources.

Secret access keys are like a password that pair with your access key ID (username). Need both to access AWS services.

Best practice

Although I'm using access keys in this project, a best practice alternative is to use AWS Cloudshell to run commands or use the AWS CLI V2 and enable authentication through a user in IAM identity center.



Connecting to my S3 bucket

The command I ran was "aws s3 ls". This command is used to list the S3 buckets in my account.

The terminal responded with the list of S3 buckets in my account. This indicated that the access keys I set up provide access to the EC2 instance to my AWS environment.

```
[ec2-user@ip-10-0-3-228 ~]$ aws s3 ls
2025-08-03 23:20:24 nextwork-vpc-endpoints-roshan0621
[ec2-user@ip-10-0-3-228 ~]$
```



Connecting to my S3 bucket

I also tested the command "aws s3 ls s3://nextwork-vpc-endpoints-roshan0621" which returned the objects inside my s3 bucket.

```
[ec2-user@ip-10-0-3-228 ~]$ aws s3 ls
2025-08-03 23:20:24 nextwork-vpc-endpoints-roshan0621
[ec2-user@ip-10-0-3-228 ~]$
[ec2-user@ip-10-0-3-228 ~]$
[ec2-user@ip-10-0-3-228 ~]$ aws s3 ls s3://nextwork-vpc-endpoints-roshan0621
2025-08-03 23:21:09      121356 AWS  Conectivity.png
2025-08-03 23:21:09      132887 AWS  Network Monitoring.png
[ec2-user@ip-10-0-3-228 ~]$
[ec2-user@ip-10-0-3-228 ~]$
[ec2-user@ip-10-0-3-228 ~]$
```



Uploading objects to S3

To upload a new file to my bucket, I first ran the command "sudo touch /tmp/test.txt". This command creates an empty text file test.txt in the directory /tmp/.

The second command I ran was "aws s3 cp /tmp/test.txt s3://nextwork-vpc-endpoints-roshan0621". This command will copy the test.txt file from the /tmp/ directory to my s3 bucket.

The third command I ran was "aws s3 ls s3://nextwork-vpc-endpoints-roshan0621" which validated that the test.txt file had been copied over to my s3 bucket.

```
[ec2-user@ip-10-0-3-228 ~]$  
[ec2-user@ip-10-0-3-228 ~]$ aws s3 ls s3://nextwork-vpc-endpoints-roshan0621  
2025-08-03 23:21:09      121356 AWS Connectivity.png  
2025-08-03 23:21:09      132887 AWS Network Monitoring.png  
[ec2-user@ip-10-0-3-228 ~]$  
[ec2-user@ip-10-0-3-228 ~]$  
[ec2-user@ip-10-0-3-228 ~]$ sudo touch /tmp/test.txt  
[ec2-user@ip-10-0-3-228 ~]$ ls /tmp/  
systemd-private-4f592bf12bf54c17afb3a937a2dcf836-chromydw.service-adm4A2  systemd-private-4f592bf12bf54c17afb3a937a2dcf836-systemd-logind.service-pe8e08  
systemd-private-4f592bf12bf54c17afb3a937a2dcf836-dbus-broker.service-FMqDQO  systemd-private-4f592bf12bf54c17afb3a937a2dcf836-systemd-resolved.service-rw6iil  
systemd-private-4f592bf12bf54c17afb3a937a2dcf836-policy-routes#ena5.service-OxuPaU  test.txt  
[ec2-user@ip-10-0-3-228 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-endpoints-roshan0621  
upload: ./../tmp/test.txt to s3://nextwork-vpc-endpoints-roshan0621/test.txt  
[ec2-user@ip-10-0-3-228 ~]$  
[ec2-user@ip-10-0-3-228 ~]$ aws s3 ls s3://nextwork-vpc-endpoints-roshan0621  
2025-08-03 23:21:09      121356 AWS Connectivity.png  
2025-08-03 23:21:09      132887 AWS Network Monitoring.png  
2025-08-03 23:38:26           0 test.txt  
[ec2-user@ip-10-0-3-228 ~]$  
[ec2-user@ip-10-0-3-228 ~]$
```



In the second part of my project...

Step 5 - Set up a Gateway

In this step, we are creating a VPC endpoint so that the VPC can communicate with S3 directly without using the public internet.

Step 6 - Bucket policies

In this step, we are limiting our S3 buckets traffic to only accept traffic coming from the VPC endpoint.

Step 7 - Update route tables

In this step, we are going to test our VPC endpoint setup and troubleshoot any connection issues.

Step 8 - Validate endpoint connection

In this step, we are going to test our VPC endpoint setup again to make sure the traffic to S3 is being routed through the endpoint instead of the internet.

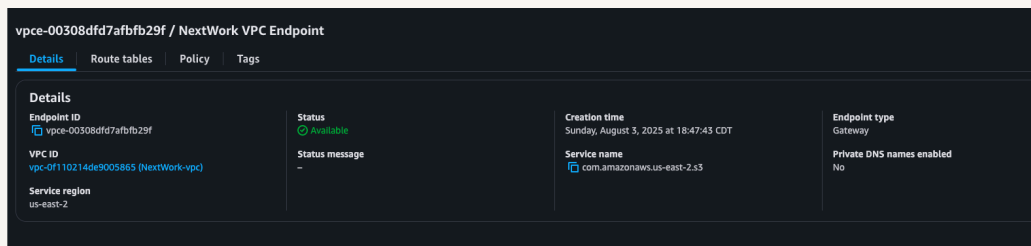


Setting up a Gateway

I set up an S3 Gateway, which is a type of endpoint used specifically for Amazon S3 and DynamoDB. This gateway will add a route to your VPC route table and direct traffic bound for S3 directly to the gateway instead of the internet.

What are endpoints?

An endpoint is a direct connection between your VPC and other AWS services without going over the internet.





Bucket policies

A bucket policy is a type of IAM policy designed to setup access permissions to an S3 bucket.

My bucket policy will deny all access to my S3 bucket except to traffic coming from my VPC endpoint.

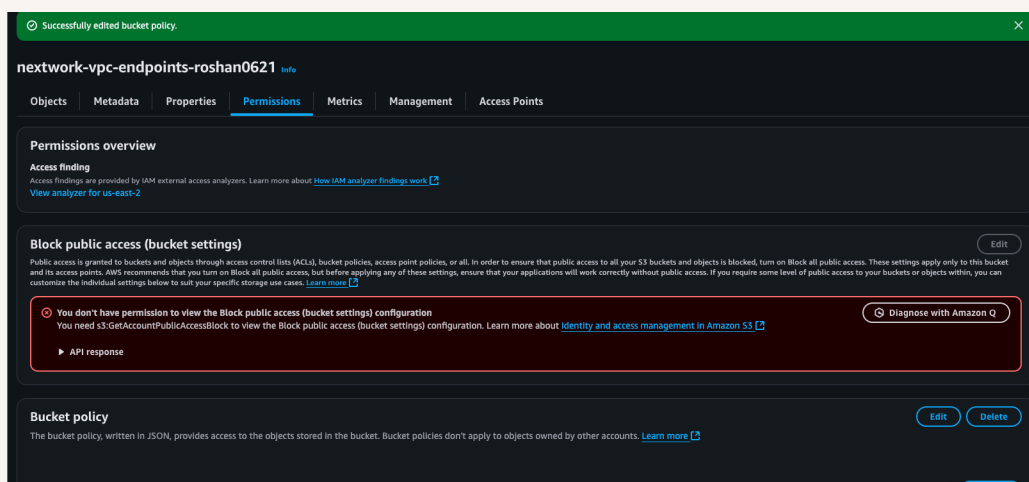
```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Deny",
6       "Principal": "*",
7       "Action": "s3:*",
8       "Resource": [
9         "arn:aws:s3:::nextwork-vpc-endpoints-roshan0621",
10        "arn:aws:s3:::nextwork-vpc-endpoints-roshan0621/*"
11      ],
12      "Condition": {
13        "StringNotEquals": {
14          "aws:sourceVpc": "vpc-00308dfd7afbfb29f"
15        }
16      }
17    }
18  ]
19 }
```



Bucket policies

Right after saving my bucket policy, my S3 bucket page showed denied access warnings. This was because the policy denies all actions unless they come from my VPC endpoint. This means even the AWS management console is denied access.

I also had to update my route table because we need to direct traffic headed to S3 specifically to go through the VPC endpoint instead of using the internet.

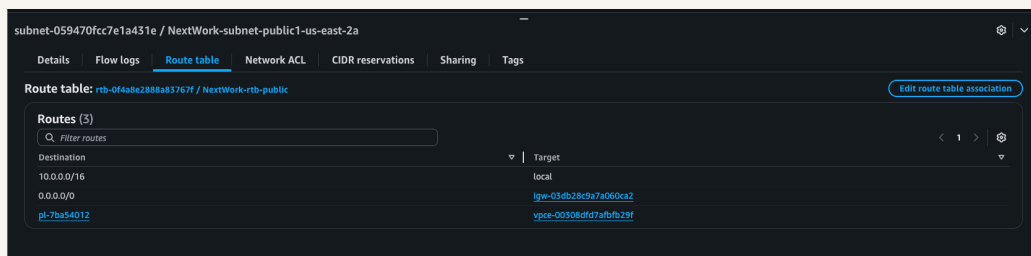




Route table updates

To update my route table, I opened my VPC endpoint, went to route tables, manage route tables, selected the VPC public route table and then modify route table.

After updating my publi subnet's route table, my terminal could return the list of objects in my S3 bucket.

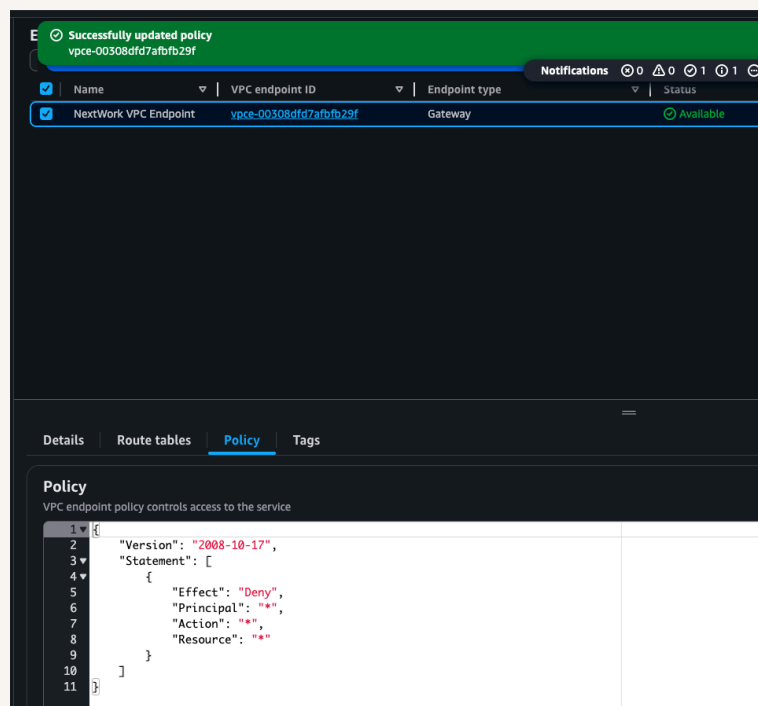




Endpoint policies

An endpoint policy is a great tool to control access to resources. Can be granular with the way access to AWS resources is given.

I updated my endpoint's policy by switching Effect from Allow to Deny. I could see the effect of this right away, because my EC2 instance could no longer access my S3 bucket via the endpoint.





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